

ZHIJIA ZHANG

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Employment

2026–2029	Columbia University Postdoctoral Research Scholar	New York, NY
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Education

2021–2026	New York University Ph.D. in Mathematics	New York, NY
2017–2021	New York University Shanghai B.S. in Honors Mathematics	Shanghai, China

Research interests

Algebraic geometry, (equivariant) birational geometry and rationality problems.

Publications

Equivariant unirationality of Fano threefolds, (with I. Cheltsov and Yu. Tschinkel),
to appear in *Annali della Scuola Normale Superiore di Pisa*, 34 pp.

Birational invariants of volume preserving maps, (with K. Loginov),
to appear in *Algebra & Number Theory*, 34 pp.

Stable equivariant birationalities of cubic and degree 14 Fano threefolds, (with Yu. Tschinkel),
to appear in *Algebraic Geometry and Physics*, 22 pp.

Cohomological obstructions to equivariant unirationality, (with Yu. Tschinkel),
International Mathematics Research Notices **2026** (15): rnag166, 16 pp.

Equivariant geometry of cubic threefolds with non-isolated singularities, (with I. Cheltsov,
L. Marquand and Yu. Tschinkel),
Pure and Applied Mathematics Quarterly **22** (2026), no. 2, 555–575.

Rationality of singular cubic threefolds over $\mathbb{R}$, (with I. Cheltsov and Yu. Tschinkel),
Advances in Mathematics **487** (2026), Paper No. 110756, 27 pp.

Equivariant geometry of singular cubic threefolds, II, (with I. Cheltsov, L. Marquand and
Yu. Tschinkel),
Journal of the London Mathematical Society **112** (2025), no. 1, e70224, 46 pp.

Computing the equivariant Brauer group, (with A. Pirutka),
International Journal of Mathematics **36** (2025), no. 8, 37 pp.

Modular symbols and equivariant birational invariants,
Journal of Number Theory **271** (2025), 308–327.

Rationality of forms of $\overline{\mathcal{M}}_{0,n}$, (with B. Hassett and Yu. Tschinkel),
Journal of the London Mathematical Society **111** (2025), no. 2, e70097, 35 pp.

Equivariant geometry of singular cubic threefolds, (with I. Cheltsov and Yu. Tschinkel),
Forum of Mathematics. Sigma **13** (2025), e9, 52 pp.

Equivariant geometry of the Segre cubic and the Burkhardt quartic, (with I. Cheltsov and Yu. Tschinkel),
Selecta Mathematica. New Series **31** (2025), no. 1, Paper No. 7, 36 pp.

Equivariant birational geometry of linear actions, (with Yu. Tschinkel and K. Yang),
EMS Surveys in Mathematical Sciences **11** (2024), no. 2, 235–276.

Combinatorial Burnside groups, (with Yu. Tschinkel and K. Yang),
Research in Number Theory **8** (2022), no. 2, Paper No. 33, 17 pp.

Preprints

Equivariant unirationality of cubic threefolds (with M. Ballard and A. Duncan), 12 pp., 2026.

Universal torsor over quartic del Pezzo surface and stable rationality (with Yu. Tschinkel),
24 pp., *submitted*, 2026.

Birational invariance of higher Amitsur groups, (with F. Scavia and Yu. Tschinkel), 28 pp.,
submitted, 2026.

Birational geometry of actions on del Pezzo surfaces, (with I. Cheltsov and Yu. Tschinkel),
46 pp., *submitted*, 2026.

Equivariant unirationality of tori in small dimensions, (with Yu. Tschinkel), 28 pp., *submitted*, 2025.

Finite abelian groups acting on rationally connected threefolds II: groups of K3 type, (with K. Loginov and A. Pinardin), 35 pp., *submitted*, 2025.

$\mathfrak{A}_5$ -equivariant geometry of quadric threefolds, (with A. Pinardin), 54 pp., *submitted*, 2025.

Conjugacy classes of linear actions in the plane Cremona group, (with I. Cheltsov and Yu. Tschinkel), 32 pp., *submitted*, 2025.

Grants and awards

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| 2026–2028 | AMS-Simons Travel Grant |
| 2025 | Charles Newman Fellowship, NYU Courant
<i>awarded each year to one or two outstanding Mathematics PhD students.</i> |
| 2024 | Wilhelm T. Magnus Memorial Prize, NYU Courant
<i>awarded for significant contributions to the Mathematical Sciences.</i> |
| 2021–2026 | MacCracken Doctoral Fellowship, NYU |

Invited talks

Seminars

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| 2026 | National Center for Theoretical Sciences, Algebraic Geometry Seminar (online talk) |
| 2026 | Leibniz University Hannover, Algebraic Geometry Oberseminar (online talk) |
| 2026 | Columbia University, Michael Zhao Memorial Student Colloquium |
| 2026 | Columbia University, Algebraic Geometry Seminar |
| 2026 | University of Utah, Algebraic Geometry Seminar |

- 2025 University of Colorado Boulder, Algebraic Geometry Seminar
- 2025 University of Illinois Chicago, Algebraic Geometry Seminar
- 2025 Columbia University, Algebraic Geometry Seminar
- 2025 University of Illinois Urbana-Champaign, Algebraic Geometry Seminar
- 2025 CUNY Graduate Center, Commutative Algebra and Algebraic Geometry Seminar
- 2025 Princeton University, Algebraic Geometry Seminar
- 2024 University of Nottingham, Pure Mathematics Seminar
- 2024 Imperial College London, Algebraic Geometry Seminar
- 2024 NYU, Algebraic Geometry Seminar

Conferences

- 2026 University of Miami, Hodge theory, Birational Geometry & Atoms
- 2026 Rutgers University, Algebraic Geometry Northeastern Series
- 2025 University of Hawaii, Polarized Varieties and Their Applications
- 2025 CIRM, Marseille, Cremona Groups
- 2025 University of Miami, New Developments of Birational Geometry
- 2024 ICMS, Edinburgh, Birational Geometry and Number Theory

Contributed presentations

- 2026 Philadelphia, International Congress of Mathematicians (short communications)
- 2025 UMass Amherst, Algebraic Geometry Northeastern Series (poster)
- 2025 CSU, Summer Research Institute in Algebraic Geometry (poster)
- 2025 UCSD, Conference on K-stability and Moduli of Varieties (short talk)
- 2024 Boston College, Algebraic Geometry Northeastern Series (short talk)
- 2023 Princeton University, Monodromy and Its Applications (short talk)
- 2022 Simons Foundation, Conference on Higher Dimensional Geometry (poster)

Teaching

New York University

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| Spring 2025 | Teaching assistant, MATH-UA.0343 Algebra I |
| Fall 2024, 2025 | Teaching assistant, MATH-UA.0348 Honors Algebra I |
| Winter 2024,
Summer 2025 | Instructor, Complex Variables Written Exam Workshop |
| Spring 2024, 2025 | Grader, MATH-GA.2140 Algebra II |
| Spring 2023, 2024 | Grader, MATH-GA.2210 Introduction to Number Theory |
| Fall 2022, 2023 | Grader, MATH-GA.2130 Algebra I |

Service

- 2025 Speaker, cSplash, NYU Courant
one-day outreach event of short lectures to local high school students
- 2024–2025 Member, PhD student survey committee, NYU Courant
- 2021–2023 Co-organizer, Graduate Student and Postdoc Seminar, NYU Courant

Refereeing: *Journal of the London Mathematical Society*, *Taiwanese Journal of Mathematics*, *Algebraic Geometry and Physics*, *EMS Surveys in Mathematical Sciences*.

Relevant skills

Languages: Chinese, English

Softwares: Magma, GAP